PMx Weekly Digest: February 21, 2026

Paper 1: Model-Based Meta-Analysis of Objective Response Rate and Survival Endpoints to C

Bibliographic Information

- **Title**: Model-Based Meta-Analysis of Objective Response Rate and Survival Endpoints to Compare PD-1 and PD-L1 Treatment Outcomes in Non-Small Cell Lung Cancer
- **Authors**: Franzese RC, Quinn L, Fava S, Roy B, Zang E, Zierhut ML, Van Der Graaf PHA
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology, 2026
- **DOI**: https://doi.org/10.1002/psp4.70196

Quick Take

A comprehensive Model-Based Meta-Analysis (MBMA) quantifying the surrogacy of ORR for OS/PFS in mNSCLC across 114 trials. It specifically addresses the debate between PD-1 and PD-L1 inhibitor efficacy, finding numerical trends favoring PD-1 but no statistical significance.

Executive Summary

This study presents a large-scale MBMA evaluating the efficacy landscape of PD-(L)1 inhibitors in metastatic non-small cell lung cancer (mNSCLC). By analyzing 114 studies, the authors developed a sequential modeling framework linking Objective Response Rate (ORR) to Progression-Free Survival (PFS) and Overall Survival (OS). A key contribution is the rigorous head-to-head simulation between PD-1 and PD-L1 inhibitors, a comparison often confounded in individual trials by patient selection. While the model predicts a numerical advantage for PD-1 inhibitors (particularly in high PD-L1 expression subgroups), the difference did not reach statistical significance. The work establishes a robust quantitative framework for benchmarking emerging therapies and simulating late-stage trial outcomes based on early-phase ORR data.

⚡ Methodological Snapshot

The authors employed a two-stage MBMA approach. First, a mixed-effects logistic regression model quantified the impact of treatment and covariates (including a non-linear PD-L1 expression function) on ORR. Second, mixed-effects semi-parametric proportional hazard models were developed for OS and PFS.

The survival models utilized digitized Kaplan-Meier curves reconstructed into monthly conditional event probabilities. Crucially, the predicted ORR from the first stage served as a regressor in the survival models, linking the endpoints. The framework accounted for non-proportional hazards in PFS (crossing curves) by implementing time-dependent baseline hazards for chemotherapy versus immunotherapy arms.

🏗 Structural Model Breakdown

The ORR model utilizes a logit link function where the log-odds of response are driven by treatment type, line of therapy, and a treatment-specific function of PD-L1 expression (linear for PD-L1 inhibitors, quadratic for PD-1 inhibitors).

The Survival models (OS/PFS) define the hazard at time t as a product of a baseline hazard $h_0(t)$ (piecewise constant monthly intervals) and an exponential function of covariates. The key covariate is the predicted ORR, with slope parameters specific to the treatment class (e.g., Mono vs. Chemo-Combo). Random effects were placed on the study-level intercept and the ORR slope.

Detailed Methodological Analysis

Modeling Approach

Sequential MBMA: Stage 1 Mixed-effects logistic regression (ORR); Stage 2 Mixed-effects semi-parametric proportional hazards (OS/PFS). Implemented in R.

Data Sources

114 clinical trials (Phase 1-3) from Certara CODEX PD-(L)1 Solid Tumor Database. Included monotherapies and combinations.

Estimation Methods

Maximum likelihood estimation for fixed effects; Random effects estimated for between-study variability (likely Laplacian or FOCE-like approximation within the R framework used).

Model Evaluation

Goodness-of-fit assessed via visual predictive checks (comparing arm-specific predictions vs observed), forest plots, and physiological plausibility of parameter estimates.

Covariate Analysis

Random forest imputation for missing covariates. Key covariates: PD-L1 expression (continuous and categorical), histology (squamous vs non-squamous), ECOG PS, race (Asian vs non-Asian), and line of therapy.

Scientific Context & Motivation

In oncology drug development, the relationship between short-term endpoints (ORR) and long-term survival (OS) is critical for accelerating approval and decision-making. While PD-(L)1 inhibitors are standard of care in mNSCLC, the concordance between ORR and OS can be inconsistent, and the comparative efficacy of PD-1 versus PD-L1 inhibitors remains a subject of debate. This study addresses these gaps by aggregating a massive dataset to quantify these relationships while adjusting for confounding covariates like PD-L1 expression, histology, and line of therapy.

Statistical Rigor Assessment

The study demonstrates high rigor by using a large sample size (114 studies) and addressing common meta-analysis pitfalls. The use of random forest for covariate imputation is robust. A notable statistical strength is the handling of non-proportional hazards in PFS (crossing curves) by allowing time-dependent baseline hazards for chemotherapy, preventing bias in Hazard Ratio estimation.

📊 Key Findings

ORR was confirmed as a significant predictor of OS and PFS across treatment classes. The model identified a quadratic relationship between PD-L1 expression and ORR for PD-1 inhibitors, suggesting enhanced efficacy at higher expression levels compared to the linear relationship seen with PD-L1 inhibitors.

Simulations of head-to-head comparisons (e.g., pembrolizumab vs. atezolizumab) showed a numerical trend favoring PD-1 inhibitors (ORR Odds Ratio ~1.39; OS HR ~0.87), but confidence intervals crossed unity (OS HR 95% CI: 0.69–1.02), indicating no statistical significance. For low PD-L1 populations, adding chemotherapy to PD-(L)1 significantly improved ORR and PFS, with a non-significant trend for OS benefit.

💡 Clinical & Regulatory Implications

The lack of statistically significant difference between PD-1 and PD-L1 inhibitors supports their interchangeable use in broad guidelines, though the numerical trend suggests PD-1 might be preferred in high PD-L1 expressors if maximizing magnitude of response is critical. The established ORR-to-OS correlation enables sponsors to use Phase 2 ORR data to probabilistically forecast Phase 3 survival outcomes, optimizing go/no-go decisions.

Strengths & Limitations

Strengths

- Large, comprehensive dataset (114 studies) covering diverse regimens.
- Sophisticated handling of crossing survival curves (non-proportional hazards) in PFS.
- Differentiation of PD-L1 expression effects (linear vs. quadratic) by drug class.
- Robust validation using leave-one-out or similar internal checks implied by the rigorous goodness-of-fit.

Limitations (Acknowledged by Authors)

- Reliance on summary-level data (potential for ecological fallacy).
- Inconsistent reporting of covariates across published trials.
- Grouping of all chemotherapy regimens into a single category due to data sparsity.

Limitations (Expert Review)

- The ‘Other’ treatment category is vague and may introduce heterogeneity.

- Lack of individual patient data prevents analysis of patient-level surrogacy (R-squared individual).
- Potential publication bias in the source studies is not explicitly modeled.

Generalizability
Highly generalizable to mNSCLC drug development; the framework is applicable to other solid tumors if data permits.

Key Equations
ORR Logistic Regression Model

$$P(\text{event})_{ij} = \text{logit}^{-1}\big(\theta_{\text{treat}} + \theta_{\text{cov}} \cdot \text{Cov}_{ij} + f(\text{PDL1}) + \eta_i\big)$$

Predicts the probability of response based on treatment, covariates, and random study effects.

Survival Hazard Model

$$h(t) = h_0(t) \cdot \exp\big(\beta_{\text{treat}} + \lambda_{\text{class}} \cdot \text{ORR}_{\text{pred}} + \gamma \cdot \text{Cov} + \eta_{\text{study}}\big)$$

Proportional hazards model linking the predicted ORR to the hazard of death or progression.

Binomial Error Model

$$\sigma_{ij}^2 = \frac{P_{ij}\big(1 - P_{ij}\big)}{N_{ij}}$$

Variance of the observed proportion (ORR) used for weighting in the meta-analysis.

Figures & Tables
- **Figure 2**: Forest plot of simulated head-to-head comparisons (Hazard Ratios) between PD-1 and PD-L1 inhibitors.
  - *Significance*: Visually demonstrates the numerical trend favoring PD-1 inhibitors while highlighting the overlap with unity (lack of statistical significance).
- **Figure 5**: Probability of success (PoS) plots for hypothetical head-to-head trials.
  - *Significance*: Provides actionable intelligence for trial design, showing that while PD-1 has a higher probability of being superior, the most likely outcome is non-inferiority.
- **Table 1**: Parameter estimates for ORR, OS, and PFS models.
  - *Significance*: Key for model reproducibility; shows the specific coefficients for PD-L1 expression and the magnitude of ORR-to-Survival translation.

Supplementary Materials
Includes detailed covariate imputation methods (Random Forest), goodness-of-fit plots, and sensitivity analyses confirming robustness to missing PD-L1 data.

Future Directions

Extension of the framework to include newer modalities like ADCs or bispecifics. Application of the methodology to individual patient data (IPD) to better characterize within-study correlations between endpoints.

Expert Commentary

This is a textbook example of how MBMA can synthesize fragmented evidence to answer questions that single trials cannot. The finding regarding the 'crossing curves' in PFS—where immunotherapy looks worse than chemo initially (pseudo-progression or delayed effect) but better long-term—is crucial. Standard Cox models often fail here, and the authors' use of time-dependent baseline hazards is the correct pharmacometric solution. The distinction between PD-1 and PD-L1 remains a subtle but persistent signal in the literature; this paper suggests that while the mechanism might differ (PD-1 blocking PD-L2 interaction), the clinical delta is small enough to be washed out by between-trial variability.

Bottom Line

For pharmacometricians in oncology, this paper provides the definitive reference parameters for linking ORR to OS in mNSCLC. It demonstrates that while PD-1 inhibitors may have a slight biological edge over PD-L1 inhibitors in high-expression tumors, the clinical outcomes are largely comparable. The methodology serves as a template for handling non-proportional hazards and covariate integration in aggregate data meta-analyses.

📊 Figures

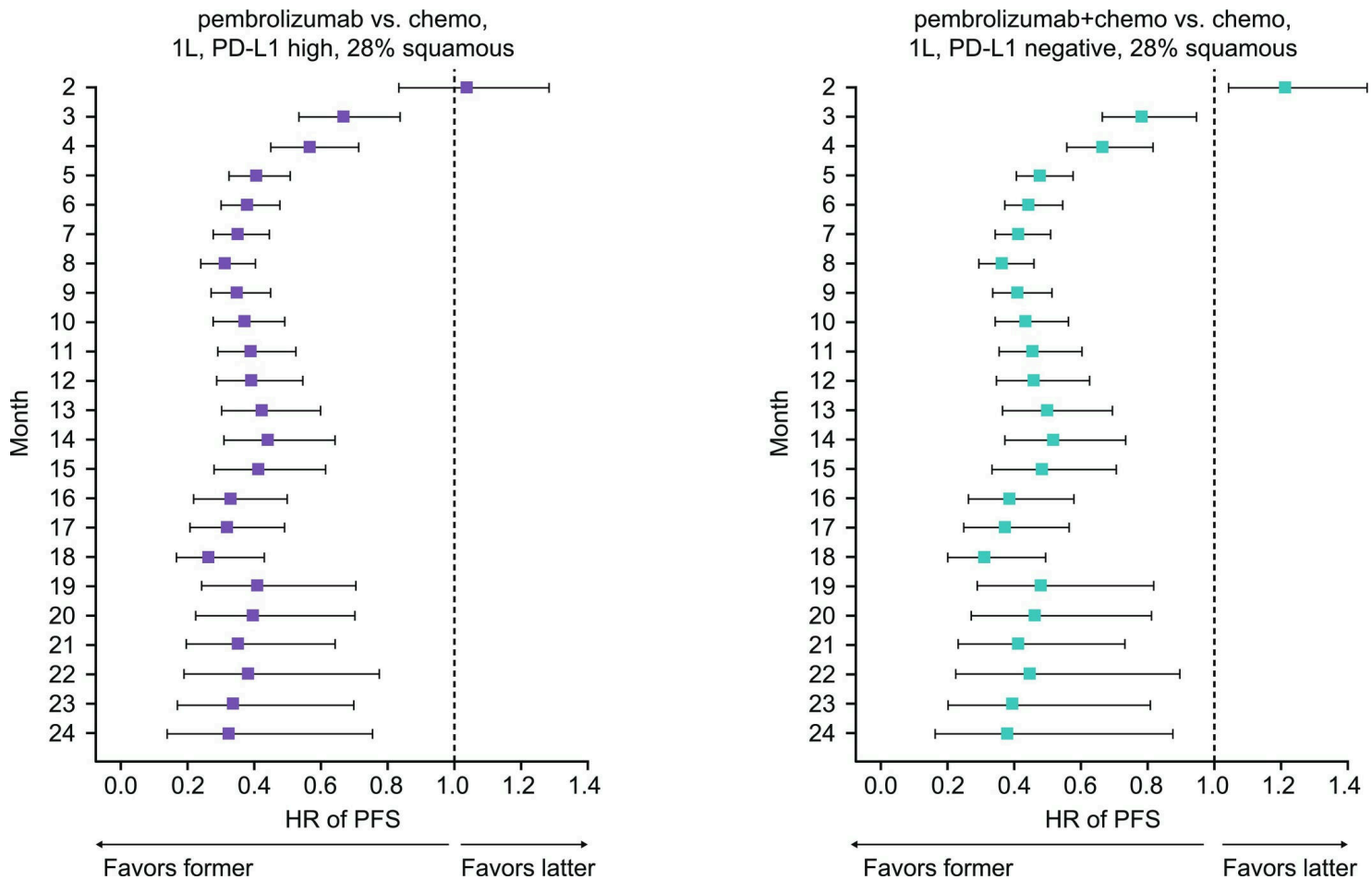

Figure 1: FIGURE 1: Predicted time-dependent PFS HR illustrated using pembrolizumab. Squares represent the median predicted HR; horizontal bars are the 95% CI of the predicted HR. 1

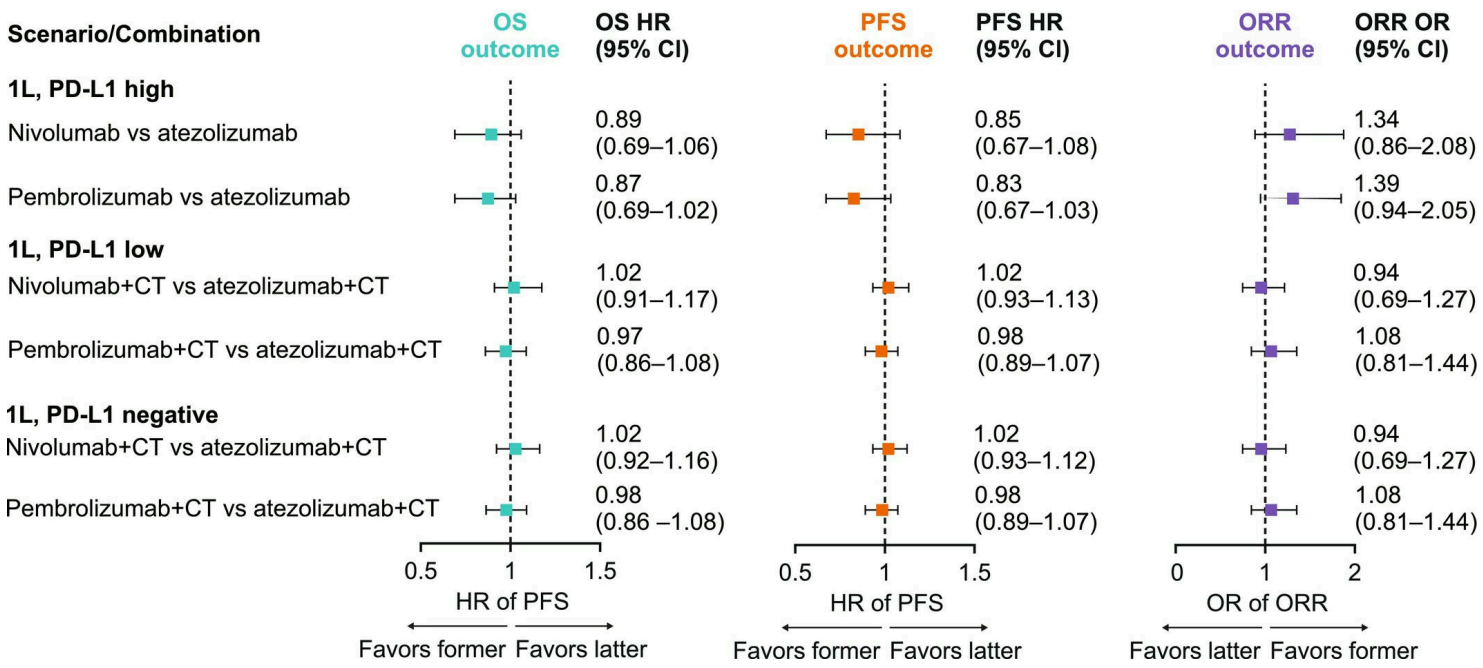

Figure 2: FIGURE 2: Model-based simulation of head-to-head trials of PD-1 versus PD-L1 inhibitors. Squares represent the median predicted HR; horizontal bars are the 95% CI of the p

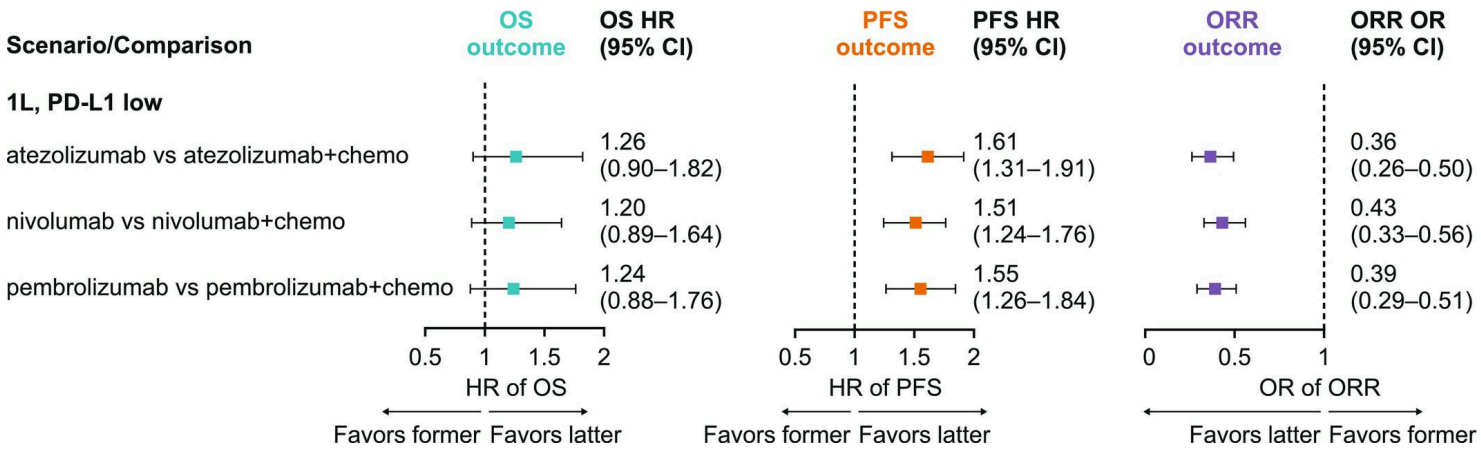

Figure 3: FIGURE 3: PD-L1 low model-based simulation of head-to-head trials of PD-1 inhibitor monotherapy versus PD-1 inhibitor + CT. Squares represent the median predicted HR; hori

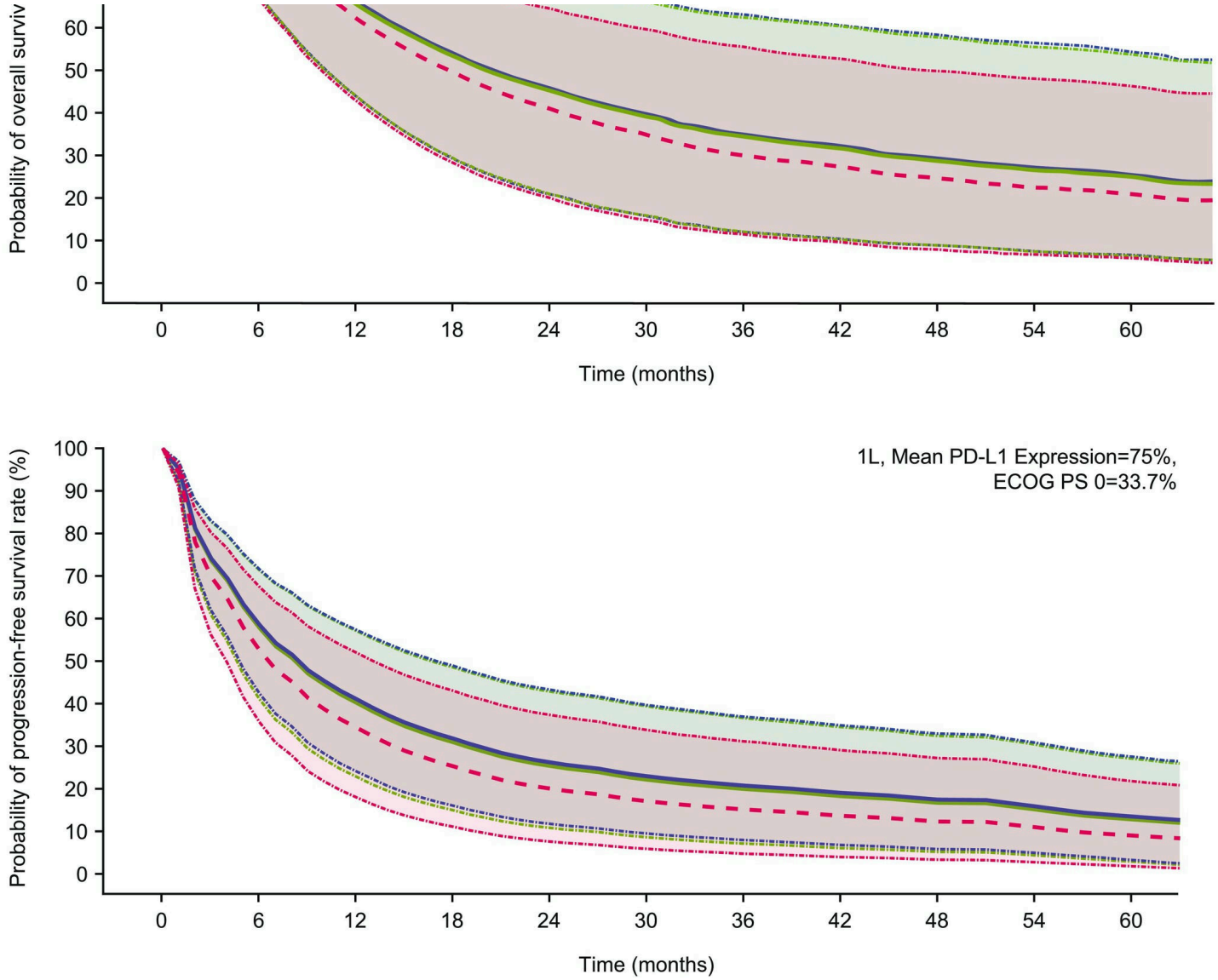

Figure 4: FIGURE 4: Joint ORR-survival simulations. 1L, first line; CI, confidence interval; CT, chemotherapy; ECOG PS, Eastern Cooperative Oncology Group Performance Status; ORR, o

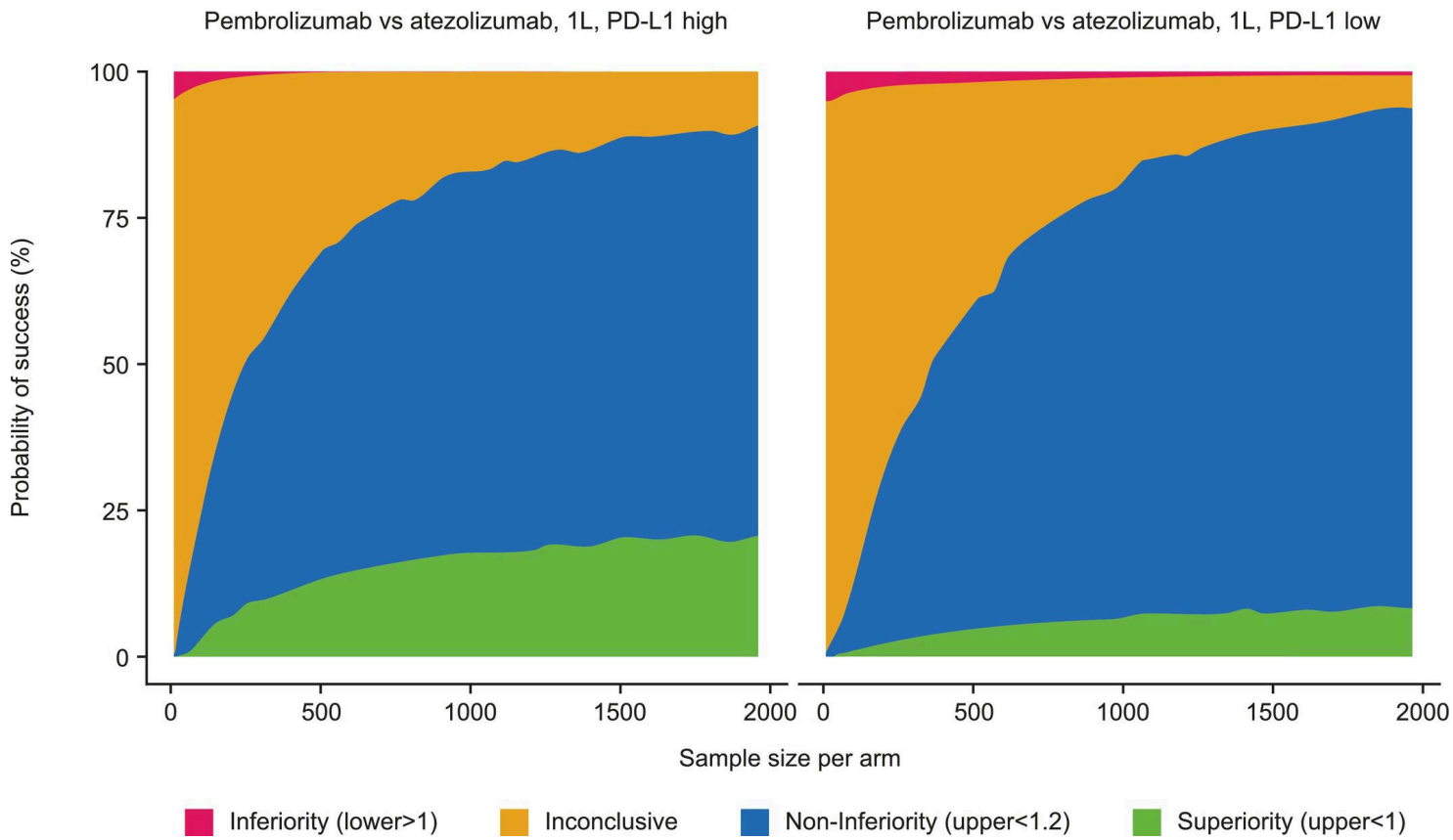

Figure 5: FIGURE 5: Probability of success calculations based on OS for head-to-head trials. 1L, first line; CI, confidence interval; lower, lower bound of a 95% CI; OS, overall sur